



## Safe mode

The F8 key is one that we only hit in the time of need, a situation where your whole system is completely messed up and you have nothing else to do but resort to the F8 key to clean up the mess. Our online lives are something like this and always connected, with Facebook and Twitter as much a part of our lives as our morning coffee. Do we really have such a fall-back in the real world? Isn't it better to be safe than sorry?

# Web Of Deceit: Everything is not as it seems

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Our online lives operate on a far greater degree of trust and ignorance than we would like to have or acknowledge. We don't "go online" anymore, we're always online and we just trust that everything is in order; our passwords safe, our transactions secure and our social networks and e-mails private. We tend to draw comfort from the sight of a green HTTPS text and the lock icon. We shouldn't.

Unfortunately, the web may be a "free" platform, but its inner workings are obscured from most users. Let's put it this way, if computers were people, they communicate with each other in a universal language, say, English. Secure connections simply talk to each other

in Chinese. All it takes is someone else who understands Chinese to figure out the conversation. This is simplifying the process a little too much, but as a basic explanation, it's enough.

### Know thine enemy

HTTPS (and the internet in general) primarily secures itself using SSL (Secure Socket Layer) / TLS (Transport Layer Security) protocols. Basically, an SSL certificate is granted to an entity once that entity's credentials have been verified by a Certificate Authority (CA). This certificate is very similar to a PAN card, driving license or any other such identifying document. When you visit a secure website, the browser verifies that the certificate issued to that website is

valid, that it was issued by an authority such as VeriSign and that this is indeed the website you were looking for.

Herein lays the problem. There are over 600 CAs distributed in over 50 countries and not one of them has a governing body. You just have to "trust" that they do the right thing and are issuing the certificate after the requisite verifications. Would you trust a driver who says he got a driving license in Timbuktoo? Yet to use the internet, you have to do just that. Some firms do maintain a very strict policy with regards to verification and these are the only firms that are really respected.

CAs can be classified into three types. Root CAs are ones that are trusted by the browser itself, CAs trusted by root